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The influence of solvent properties and biomass feedstock characteristics on deep eutectic solvent mediated lignin and hemicellulose fractionation

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ABSTRACT

Deep eutectic solvents (DESs) have received attention due to their ability to fractionate lignin and hemicelluloses from lignocellulosic biomass, though the lack of standard methods to predict the performance of DESs is a concern. The Kamlet-Taft solvatochromic parameters quantify hydrogen bonding and polarity properties of DESs and have been used to draw broad structure-reactivity relationships. This analysis has not been applied to similarly structured DESs despite known differences in treatment performance. Additionally, examination of structure-reactivity trends between different feedstocks has not been considered. Herein, DESs composed of choline chloride and monocarboxylic acids were employed in fractionation experiments with both hardwood and softwood biomass sources aiming to correlate fractionation performance with Kamlet-Taft parameters. Solubilized lignin was recovered and analyzed via 2D-HSQC Nuclear Magnetic Resonance (NMR) and Gel Permeation Chromatography (GPC). Delignification correlated with decreased basicity, showing predictability for both feedstock types. Arabinan and mannan removal was correlated with increasing net acidity and decreasing basicity as increased acidity enhanced hydrolysis of glycosidic linkages in galactoglucomannans and arabinoxylans. This work evaluates the Kamlet-Taft parameters as a tool to screen DESs for fractionation performance across different feedstocks and an investigation of the factors which contribute to differences in lignin removal between feedstocks.

1. Introduction

Deep eutectic solvents (DESs) have recently emerged as a promising class of solvent suited for dissolution of lignocellulose due to their advantageous physicochemical properties including, low toxicity, ease of preparation, low volatility, and chemical/thermal stability [1]. DESs are defined as mixtures of two or more compounds denoted as hydrogen bond donors (HBDs) and acceptors (HBAs), such that the eutectic point melting temperature of the mixture is significantly below what would be predicted assuming an ideal liquid mixture [2]. Quaternary ammonium salts are common choices for HBAs, whereas different compounds including carboxylic acids, amides, and alcohols have been utilized as HBDs, thereby offering a high degree of tunability afforded by different combinations of these compounds [1]. Previous work has demonstrated the potential of DESs to effectively solubilize both lignin and xylan from biomass samples, leaving glucan largely intact [3–5]. This selective fractionation behavior is ideal for applications such as dissolving pulp production and biomass pretreatment for enzymatic saccharification processes [6–8]. Moreover, DESs offer an environmentally friendly alternative to conventional biomass conversion methods; DESs prepared from bio-based components have shown favorable environmental impacts in life cycle assessments [9–12]. The present work focuses on DESs as a means of treating biomass samples by removing hemicelluloses and lignin while conserving cellulose for downstream processing.

Recent research has shifted towards developing rational design principles for specific applications, driven by the vast number of possible chemical combinations [13,14]. Current DES screening methodologies often involve measuring solvent properties, such as Hansen solubility parameters or Kamlet-Taft solvatochromic parameters, to explain performance variations. These parameters are then correlated with DES treatment performance metrics, such as the solubility of lignin and xylan, for targeted fractionation from lignocellulose. DESs solubilize lignin primarily via hydrolysis of β-O-4 ether linkages between units [3]. Similarly, hemicelluloses such as xylan are extracted through hydrolysis of glycosidic linkages between saccharide repeating units [15]. The Kamlet-Taft solvatochromic parameters, namely the hydrogen bond donating ability/Lewis acidity (α), hydrogen bond accepting ability/Lewis basicity (β), dipolarity (π*), and electronic transition energy (E_T),

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have shown some success in correlating with DES treatment performance metrics [16].

The alkyl chain length, number of functional groups, and type of functional groups present in both the HBA and HBD have been shown to have a considerable effect on the solvent properties of the DES, with shorter alkyl chain lengths and the presence of hydroxyl groups favoring increased reactivity [17–20]. Fractionation experiments with lignocellulosic substrates as well as the solubility of isolated biomass components in selected DESs have been used to correlate the Kamlet-Taft parameters with DES treatment performance metrics and biomass solubility, respectively. Oh et al. [21] prepared 25 choline chloride-based DESs using HBDs with carboxyl, hydroxyl, and amide functional groups. Treatment of pine wood revealed that increased acidity and dipolarity led to enhanced lignin removal. A similar study of lactic acid-based DESs with different HBAs showed that both lignin and xylan removal were favored by increased acidity [20]. Zhang et al. [5] correlated the Kamlet-Taft parameters of 56 DESs with the solubility of isolated lignin and xylan and reported that xylan solubility was not well explained by the acidity but by increased basicity, which displayed positive linear behavior. Lignin solubility could be well explained by a derived parameter known as the net acidity, acquired by taking the difference between the α and β parameters. Similar results were obtained in an investigation of dealkiline lignin, sodium lignosulfonate, and enzymatic hydrolysis lignin dissolved in lactic acid-based DESs which demonstrated a positive correlation between solubility and α-β [22]. However, this has yet to be shown via direct fractionation of native biomass.

In many cases, DESs with significantly different structures and properties were chosen for analysis in order to provide a broad distribution of parameter values [5,21]. This approach emphasizes differences in performance between DESs with distinct hydrogen bond donating and accepting behaviors and results in broad performance trends among the selected DESs. However, experiments using DESs with similar structures and properties have nonetheless shown significant differences in fractionation performance [3,20], a phenomenon that has not been fully explained via the Kamlet-Taft methodology. Yu et al. [23] measured the solubility of kraft lignin in carboxylic acid-based DESs with similar properties and found that relatively small differences in α values between the DESs corresponded with differences in lignin dissolution. Fractionation experiments must be carried out to confirm whether lignin removal from biomass by similarly structured can be adequately explained by the Kamlet-Taft methodology.

Another factor which has yet to be fully explored in the context of the Kamlet-Taft methodology is the type of biomass feedstock used for DES treatment. Lignocellulose is derived from a diverse range of woody and non-woody plant species with varying chemical compositions and structures. The differences between lignin types in hardwood and softwood species are well known [24]. Lignin in softwoods is predominantly comprised of guaiacyl (G) units which do not have a methoxy group at the C-5 position, allowing for greater degrees of branching and cross linking. Softwood species are also known to contain higher lignin contents than hardwoods [25]. In hardwoods, both guaiacyl and syringyl (S) units, which contain methoxy groups at both ortho-positions, are present, resulting in a more linear structure with greater β-O-4 content. Comparison of DES fractionation between Douglas fir and poplar wood showed that delignification was more pronounced for poplar than for fir at optimal conditions [3]. It has been suggested that hardwood lignin is more susceptible to DES attack due to the high β-O-4 content imparted by S units, resulting in selective cleavage of S units by the DES. Further study is required to confirm whether this is the case. The major hemicelluloses found in hardwoods and softwoods also differ, with glucuronoxylan dominating in hardwood while softwoods contain more galactoglucomannan [26]. The Kamlet-Taft methodology has been applied to xylan removal in several cases [5,20,27], whereas analysis of arabinan and mannan removal is lacking. A more complete understanding of DES-hemicellulose interactions is important for applications in which hemicellulose removal is desired.

This research aims to provide a more complete understanding of the relationship between the Kamlet-Taft parameters of a DES and its ability to separate lignin and hemicellulose from lignocellulosic biomass. The Kamlet-Taft parameters of DESs composed of choline chloride and five carboxylic acids are determined. Fractionation experiments are performed with both hardwood and softwood feedstocks to directly compare their performance. Correlations between DES Kamlet-Taft parameters are drawn for both feedstocks in a novel comparison, evaluating the wider applicability of the Kamlet-Taft methodology to multiple feedstock types. Lignin recovered from spent liquor is analyzed via 2D-HSQC NMR and GPC to further investigate differences in performance between the two feedstocks.

2. Experimental

2.1. Materials

Choline Chloride, formic acid, acetic acid, propionic acid, butyric acid, valeric acid, and lactic acid, N,N-diethyl-4-nitroanaline, 4-nitroaniline, and Nile Red were obtained from Fisher Scientific. All chemicals were of analytical purity and were used as received unless otherwise specified. Aspen (*Populus tremuloides*) and loblolly pine (*Pinus taeda*) wood chips were used as feedstocks for fractionation experiments.

2.2. Measurement of Kamlet-Taft parameters

The Kamlet-Taft solvatochromic parameters of the five selected DESs were determined based on the average of three UV–vis absorption experiments using solutions of three indicator dyes dissolved in the DES samples. N,N-Diethyl-4-nitroanailine (DENA), 4-nitroaniline (4NA), and Nile Red (NR) were employed for solvatochromic measurements. The method was first validated by measuring the Kamlet-Taft parameters of methanol, a solvent for which literature values are widely available. The same dye probes and dye concentrations were used for all measurements to ensure comparability between DESs. DES molar ratios were chosen based on the long-term stability of the DESs at room temperature.

The three dye probes were dissolved in methanol at a concentration of 1 mg/mL. Next, an aliquot of 20 μL of each dye solution was pipetted into 3 mL of each DES. The DES loaded with dyes were vortex mixed until homogeneous and then transferred into plastic cuvettes for UV–vis analysis at room temperature. The Kamlet-Taft parameters of each DES were determined by measuring the wavelength of maximum absorbance (λ_{max}). These were used to calculate the hydrogen bond donating ability/Lewis acidity (α), hydrogen bond accepting ability/Lewis basicity (β), dipolarity (π*), and electronic transition energy based on nile red (E_T(NR) according to the following Eqs. [28]:

$$E_T(NR)\left(kcal \bullet mol^{-1}\right) = \frac{28591.44}{\lambda_{max}} \tag{1}$$

$$\alpha = \frac{19.9657 - 1.0241\pi^* - \upsilon_{NR}}{1.6078} \tag{2}$$

$$\beta = \frac{1.035\upsilon_{DENA} + 2.64 - \upsilon_{4NA}}{2.80} \tag{3}$$

$$\pi^* = 0.314(27.52 - \upsilon_{DENA}) \tag{4}$$

$$\upsilon = 1/\left(\lambda_{max} \times 10^{-4}\right) \tag{5}$$

where λ_{max} is the wavelength where maximum absorption was observed, υ_{NR}, υ_{DENA}, and υ_{NR} are the wave numbers derived from λ_{max} of Nile red, N,N-diethyl-4-nitroanaline, and 4-nitroaniline, respectively.

2.3. DES preparation and fractionation

The DESs were prepared by mixing the different carboxylic acids with choline chloride at the minimum molar ratio of the acid component which allowed for the mixture to remain in the liquid phase at room temperature (Table 1). The mixtures were then heated from 60 to 80 °C for 60–120 min in a silicon oil bath with magnetic stirring until a clear, homogeneous liquid was produced. The DESs were then stored in sealed bottles prior to utilization to avoid excess moisture contamination.

DES treatment experiments were carried out using both aspen and pine, each milled to pass a 0.5 mm mesh. In a typical experiment, 2.0 g of oven dried wood powders were combined with the different DES's in a 500 mL round bottom flask at a 1:10 (*w*/w) wood to liquor ratio and treated at 150 °C for 3 h in a silicon oil bath with magnetic stirring. After 3 h elapsed, the reaction mixture was cooled to room temperature using a water bath. The pulp was separated from the cooled reaction mixture via antisolvent addition; excess ethanol was introduced to the reaction mixture to act as a thinning agent, enabling vacuum filtration. The solid fraction, containing primarily cellulose, was then oven dried at 105 °C overnight. This solid fraction recovered after DES treatment is hereafter referred to as pulp. The filtrate, containing remaining DES, ethanol, as well as solubilized lignin and hemicelluloses, was reserved for further analysis.

The composition of wood feedstocks and the DES pulps were determined according to the protocol outlined by the US National Renewable Energy Laboratory [29]. Polysaccharide content was quantified after two-stage acid hydrolysis with 72 % and 4 % H_2SO_4, respectively using an Agilent 1200 HPLC equipped with a Shodex SP0810 column. Hydrolysates were recovered via vacuum filtration using glass filtering crucibles with 10–15 μm pore sizes. The acid soluble lignin was quantified from the hydrolysate using a Thermoscientific Genesys 50 UV–Visible Spectrophotometer and acid insoluble lignin was quantified gravimetrically. Removal and retention of cellulose, lignin, and hemicelluloses were calculated according to eqs. (6) and (7), respectively [23]:

$$\textit{Component Removal}\ (\%) = \frac{C_0\ (\%) - \left[\frac{\textit{Yield}\ (\%)}{100} \times C_p\ (\%)\right]}{C_0\ (\%)} \times 100 \tag{6}$$

$$\textit{Component Retention}\ (\%) = 100 - \textit{Component Removal}\ (\%) \tag{7}$$

where C_0 and C_p are the composition of the component in the feedstock and pulp, respectively and the yield refers to the yield of pulp from the input mass of the feedstock on an oven-dried basis.

2.4. Analysis of recovered fractioned lignin portions

To understand what structures of lignin were fractionated in DES pulping, the pulping filtrate of each DES and feedstock was analyzed to understand which moieties of lignin were solubilized during pulping. CC:FA was chosen as the DES for further experimentation due to its effective lignin fractionation with both feedstocks in previous experiments. Lignin recovered from initial pulping experiments at 150 °C showed evidence of significant condensation, rendering it sparingly soluble in deuterated solvents after acetylation (**Table S5**). As a result, the 2D-HSQC NMR spectra of these lignins exhibited poor signal strength and resolution (**Fig. S3**). Pulping conditions were modified to address this issue. Mild fractionation experiments were carried out using CC:FA at 130 °C for 18 h, after which antisolvent addition and solid-liquid-separation was performed as outlined above. The liquid fraction from pulping containing primarily ethanol, lignin, and residual DES was transferred to centrifuge tubes and DI water was added at a 2:1 ratio *v*/v. After 10 min of centrifugation at 4400 rpm, the lignin precipitated and could be recovered after decanting the supernatant. The precipitated lignin was dried under vacuum at 35 °C for 48 h before being ground to a powder in a mortar and pestle. The lignins recovered from the filtrate produced from the mild pulping experiments exhibited greater solubility, allowing for characterization.

The recovered lignins were acetylated according to previously established methods [30] using a 1:1 *v*/v pyridine/acetic anhydride at 80 °C for 24 h. After returning to room temperature, the reaction mixture was precipitated in 0.10 M HCl solution, and the precipitate was washed with HCl solution and water before being vacuum filtered and air dried. 2D-HSQC NMR was employed to characterize the inter-unit linkages of the precipitated lignin samples. A quantity of 60 mg of each dried lignin was dissolved in 750 μL of DMSO-d_6 before being transferred to an NMR sample tube. Analysis of lignins recovered from the pulping filtrate was carried out using a Bruker 400 MHz NMR with the pulse program hsqcetgpsp.2, a relaxation delay of 1 s, 512 blocks and 80 scans per block. Lignin NMR spectra were assigned according to previous reports [8,31–35]. GPC analysis was also performed to evaluate the molecular weight distributions of the lignin samples. The acetylated lignins were dissolved in tetrahydrofuran at 1.0 mg/mL and passed through a 0.022 mm PVDF filter before being transferred to an appropriate autosampler vial. Analysis was carried out using an Agilent Infinity II GPC with Styragel HR 4E and HR 1 columns and detection at 280 nm. Polystyrene calibration standards were employed for molecular weight determination.

Table 1
DESs prepared and their abbreviations.

Hydrogen Bond Acceptor	Hydrogen Bond Donor	Abbreviation	Molar Ratio
Choline Chloride	Formic Acid	CC:FA	1:2
	Acetic Acid	CC:AA	1:2
	Propionic Acid	CC:PA	1:3
	Butyric Acid	CC:BA	1:5
	Valeric Acid	CC:VA	1:5

3. Results and discussion

3.1. DES Kamlet-Taft solvatochromic parameters

The Kamlet-Taft parameters of the prepared DESs are shown in Fig. 1 and tabulated in **Table S1**. The hydrogen bond acidity (α) quantifies the hydrogen bond donating ability of a DES. In monocarboxylic acids, the alkyl chain acts as an electron donating group which tends to increase electron density of the oxygen in the O—H bond of the acid. This, in turn, enhances the strength of the O—H bond and reduces its ability to be deprotonated [18]. Consequently, increased alkyl chain length of the HBD causes a decrease in the DES' ability to donate protons, hence the observed decrease in α and the corresponding increase in β. The

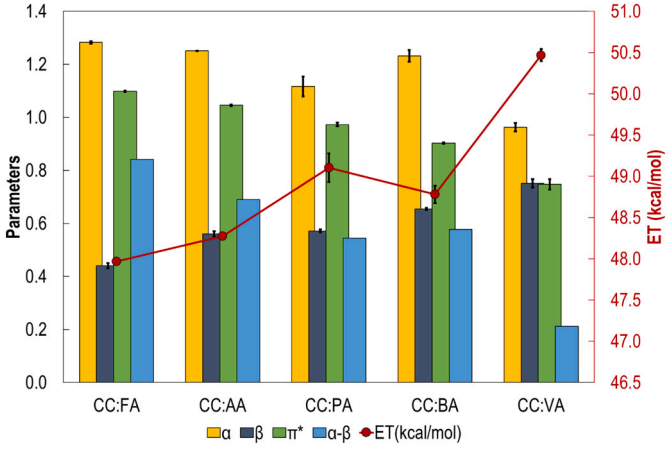


Fig. 1. Kamlet-Taft parameters of prepared DESs measured at room temperature. Error bars represent standard deviations of triplicate measurements.

hydrogen bond donating ability (β) typically exhibits an inverse correlation with α [36]. The decrease in π*, the dipolarity of the solvent, can be partly explained by this effect, though the non-polar nature of the alkyl chain moiety likely played a larger role. The electronic transition energy (E_T) is another indicator of solvent polarity similar to the Kamlet-Taft π* for which lower values are associated with higher solvent polarity. The E_T increased with increased alkyl chain length which points to reduced solvent polarity [37]. The E_T value for CC:BA was lower than that of CC:PA, indicating greater polarity despite the longer alkyl chain length of butyric acid than propionic acid. This was attributed to the presence of delocalized bonds and permanent dipoles in the solvent, allowing for greater polarity [27,38]. Positive net acidity values, taking the difference between the α and β parameters, were found for all 5 DESs meaning that each DES should be considered acidic in nature [5]. Similar to the α, the net acidity also decreased with increasing alkyl chain length.

The Kamlet-Taft parameters of CC:PA at a 1:2 M ratio have been reported in the literature and are similar to those reported here for CC:PA (1:3) [21]. Kamlet-Taft parameters have not been reported for CC:BA and CC:VA at a 1:5 M ratio, preventing direct comparison. Dwamena and Raynie [28] measured the Kamlet-Taft parameters of CC:BA and CC:VA at a 1:3 M ratio and reported values which differ from those measured in this work. In particular, the α values of CC:BA and CC:VA in the present work were elevated and the β values were lower compared to the literature values, most likely due to the greater acid content, though the π* values were in agreement (**Table S2**). The greater proportion of acid used in the present work would allow for greater availability of protons in the solvent, thereby increasing the α values. This, in turn, resulted in the lower β values as α and β are inversely correlated to each other [36]. The measurements for CC:BA and CC:VA were conducted at 40 °C compared to room temperature in this work which is another source of variance [28]. The similarity between the measured Kamlet-Taft parameter values and those reported in the literature gives confidence in the validity of the presented results.

3.2. DES pulping experiments

Pulping trials were conducted for both aspen and pine wood feedstocks with each DES and the pulps and feedstocks were analyzed via biomass compositional analysis (Table 2). The solids yield ranged from 45.6 %–62.3 % for aspen and 53.6 %–70.3 % for pine. With the exception of the CC:FA aspen pulp, these yields were elevated when compared to typical kraft and soda pulping yields of 40–50 % [39]. For the pine pulps in particular, the higher yields were likely due to lower overall extraction of lignin and hemicelluloses. The recalcitrant lignin and hemicelluloses remained within the solid fraction rather than being solubilized into the spent pulping liquor and therefore increased the pulping yield. DES treatment resulted in an increase in glucan concentration within the pulp compared to that of the feedstock, rising from 43.1 % in the untreated aspen wood to a maximum of 77.8 % for the CC:FA pulp. The glucan concentration increased resulting from the pine pulping trials, however this effect was less pronounced, rising from 36.4 % to a maximum of 53.8 % for the CC:FA pulp and a minimum of 42.7 % for CC:VA.

Xylan, galactan, and the sum of arabinan and mannan also exhibited changes in concentration due to DES pulping. In the aspen pulps, the xylan concentration was reduced significantly and the arabinan and mannan concentration was also reduced, though to a lesser extent. The galactan concentration increased slightly, likely due to the low initial concentration, and therefore availability, in the untreated aspen. For the pine pulps, the concentrations of all three hemicelluloses were reduced, an effect which was most prominent for arabinan and mannan. This is due to the prevalence of galactoglucomannans in softwoods. The lignin concentration of both aspen and pine was reduced as a consequence of DES pulping. Once again, this reduction was more pronounced for the aspen pulps, falling from 21.6 % to a minimum of 5.47 % in the CC:FA pulp. The concentration of lignin in the pine pulps was reduced from 32.6 % to a minimum of 17.4 % for CC:AA, though CC:BA and CC:VA reached only 28.1 % and 29.5 %, respectively. Ash content was largely unaffected by DES pulping.

The changes in the concentrations of lignin, cellulose, and hemicelluloses were further quantified according to Eq. 6. CC:FA exhibited the best performance for aspen with lignin and hemicellulose removal of 88.3 % and 79.5 % respectively and 82.9 % cellulose retention (Fig. 2a). In contrast, CC:VA achieved only 69.4 % lignin separation and 66.7 % hemicellulose removal; however, CC:VA showed the highest cellulose retention of 91.0 %. In fact, while fractionation of lignin and hemicellulose decreased from CC:FA to CC:VA, cellulose retention increased slightly. Though it should be noted that cellulose retention was high for all DES across both feedstocks, showing that potential hydrolysis of cellulose chains was limited during pulping trials. DES treatment of pine wood was overall less successful than aspen, with delignification ranging from 39.9 %–67.6 % and hemicellulose removal ranging from 48.9 %–73.9 % (Fig. 2b). Variability in cellulose retention was increased compared to aspen, ranging from 66.0 %–95.9 %. The ability of the DESs to extract lignin from pine was severely inhibited, especially for CC:BA and CC:VA. In this manner, DES pulping bears similarity to Kraft pulping which is also known to underperform with softwood feedstocks compared to hardwoods [24].

Lignin removal exhibited a strong negative correlation with the solids yield for the aspen pulping trials, though no correlation could be drawn for the pine pulping trials (Fig. 2c). In the case of the pine pulping trials, lignin removal was not correlated with the solids yield. Lower lignin removal results in a greater fraction of lignin remaining in the

Table 2
Composition of untreated aspen and pine wood and pulps after fractionation with each DES.

Sample	Solids Yield[a] (%)	Glucan (%)	Xylan (%)	Galactan (%)	Arabinan + Mannan (%)	Lignin (%)	Ash (%)	Mass Balance[b] (%)
Aspen	–	43.1 ± 0.3[c]	23.0 ± 1.2	1.2 ± 0.4	4.7 ± 0.3	21.6 ± 0.2	0.21 ± 0.02	93.8
CC:FA	45.9	77.8 ± 0.5	8.1 ± 0.4	2.3 ± 0.1	2.6 ± 0.3	5.47 ± 0.3	0.27 ± 0.05	96.4
CC:AA	55.5	68.8 ± 0.5	8.3 ± 0.4	2.7 ± 0.2	3.8 ± 0.6	8.52 ± 0.9	0.25 ± 0.04	92.3
CC:PA	56.3	67.1 ± 0.5	10.7 ± 0.6	1.9 ± 0.1	3.9 ± 0.5	7.88 ± 1.0	0.22 ± 0.02	91.7
CC:BA	57.4	65.8 ± 0.5	11.9 ± 0.6	2.0 ± 0.1	3.9 ± 0.5	8.15 ± 1.5	0.22 ± 0.02	92.0
CC:VA	62.3	63.0 ± 2.5	9.6 ± 0.5	1.9 ± 0.1	4.0 ± 1.1	10.6 ± 1.5	0.22 ± 0.04	89.3
Pine	–	36.4 ± 0.3	10.5 ± 0.6	3.7 ± 0.2	15.1 ± 0.2	32.6 ± 0.1	0.20 ± 0.01	98.5
CC:FA	55.9	53.8 ± 1.9	5.8 ± 0.9	1.7 ± 0.2	6.1 ± 0.5	22.8 ± 2.2	0.25 ± 0.01	90.5
CC:AA	60.5	50.9 ± 1.3	8.3 ± 1.4	2.2 ± 0.3	9.3 ± 0.6	17.4 ± 1.0	0.26 ± 0.02	88.4
CC:PA	70.3	49.7 ± 0.8	8.8 ± 1.7	2.2 ± 0.3	10.3 ± 0.2	19.2 ± 0.9	0.26 ± 0.02	90.4
CC:BA	53.6	44.8 ± 2.1	6.8 ± 1.4	1.6 ± 0.3	10.6 ± 0.5	28.1 ± 0.7	0.26 ± 0.02	92.1
CC:VA	66.4	42.7 ± 2.0	7.2 ± 1.5	2.0 ± 0.3	11.0 ± 0.3	29.5 ± 0.01	0.26 ± 0.05	92.5

[a] Solids yield after DES treatment.
[b] Sum of biomass components determined by compositional analysis.
[c] Standard deviation.

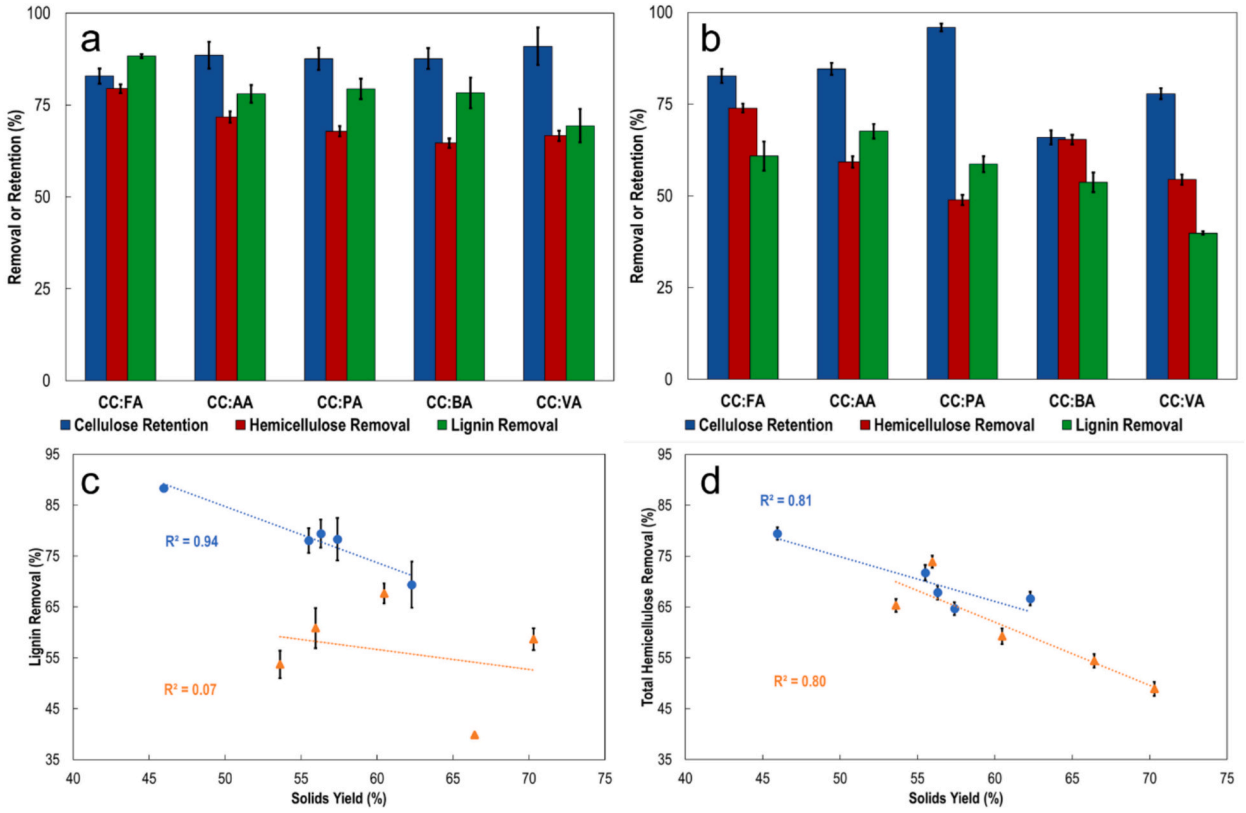


Fig. 2. Cellulose retention, hemicellulose removal, and lignin removal for aspen (a) and pine (b) wood after DES treatment. Relationship between solids yield after DES treatment and lignin removal (c) and total hemicellulose removal (d) from aspen (blue) and pine (orange).

pulp which causes the higher pulping yield, an effect which has been reported elsewhere [18]. The poor correlation seen for the pine pulps may be due to the comparatively greater variability in hemicellulose removal and cellulose retention. Total hemicellulose removal trended negatively with the solids yield for both feedstocks with strong correlation (Fig. 2d). This indicates that DES-hemicellulose interactions played a major role during the DES treatment process, especially for the pine feedstock.

3.3. Lignin removal

The primary mechanism of delignification during acidic DES treatment has been suggested to proceed via cleavage of β-O-4 ether linkages between lignin units initiated by protonation of the hydroxyl group on the α-carbon of the lignin sidechain [3,23,40,41]. DESs with greater acidity have a greater capacity to donate protons in solvent-solute interactions, likely allowing for greater lignin solubilization and separation from the biomass substrate. A study of the solubility of kraft lignin in carboxylic acid based DESs found that lignin solubility was well correlated with increasing α and π* and, to a lesser extent, increasing β [23]. This confirmed that increased acidity and polarity enhanced the reactivity of the DESs with lignin. Further corroboration of this effect was found by treatment of corn stover with lactic acid based DESs which demonstrated a positive correlation between lignin removal and α [20].

A positive relationship between lignin removal and α was found (Fig. 3a). CC:FA and CC:AA, which showed the highest α values, achieved the highest lignin removal for both feedstocks. Similarly, CC:VA showed the lowest α value and the lowest lignin removal. The net acidity was also well correlated with lignin removal (Fig. 3b), exhibiting good agreement with kraft lignin solubility experiments which demonstrated a positive relationship between lignin solubility and α-β [5]. The dipolarity also exhibited a positive correlation with lignin removal (Fig. 3c). This can be explained by the relationship between the polarity and acidity of the DESs. As discussed, CC:FA has a greater α value than CC:VA because the increased alkyl chain length of valeric acid stabilizes the acidic O—H bond of the carboxyl group. The shorter alkyl chain length of formic acid does not provide this stabilizing effect, resulting in the increased proton donating ability of CC:FA. The stabilization of the acidic O—H bond by the alkyl chain acts by reducing the polarity of the bond. As a result, lignin removal is positively correlated with both acidity and dipolarity. Finally, a negative correlation with β was observed (Fig. 3d), recreating a relationship derived from kraft lignin solubility measurements [5]. The Kamlet-Taft parameters demonstrated a strong capability to explain differences in lignin removal between the selected DESs. Importantly, the observed correlations were found for both aspen and pine, providing evidence for the first time that the Kamlet-Taft methodology can be applied to different feedstocks for DESs composed of choline chloride and carboxylic acids.

3.4. Hemicellulose removal

Efficient xylan fractionation was observed for both aspen and pine, ranging between 70 and 80 % for all 5 DESs and both feedstocks. Previous analyses of xylan solubility in DESs have predicted that DESs with greater β values have a greater ability to solvate xylan [5,27]. According to these findings, the DES interacts with xylan by aggregating at the DES-xylan interface and accepting hydrogen bonds from hydroxyl groups on the xylan chain. The xylan is then deprotonated by the DES which enhances its solubility, allowing for dissolution of up to 60 % of xylan [27,42,43]. However, this is true only for DESs with strong hydrogen bond accepting abilities which typically have β values greater than 1.0. In fact, the DESs used by Yu et al. [27] to establish positive correlations between xylan solubility and β showed β values ranging from 1.2 to 1.7, indicating strong basicity. The DESs in the present work were acidic; their acidity is evidenced by their positive net acidity and much lower β values which ranged from 0.4 to 0.8. Therefore, the DESs in this study

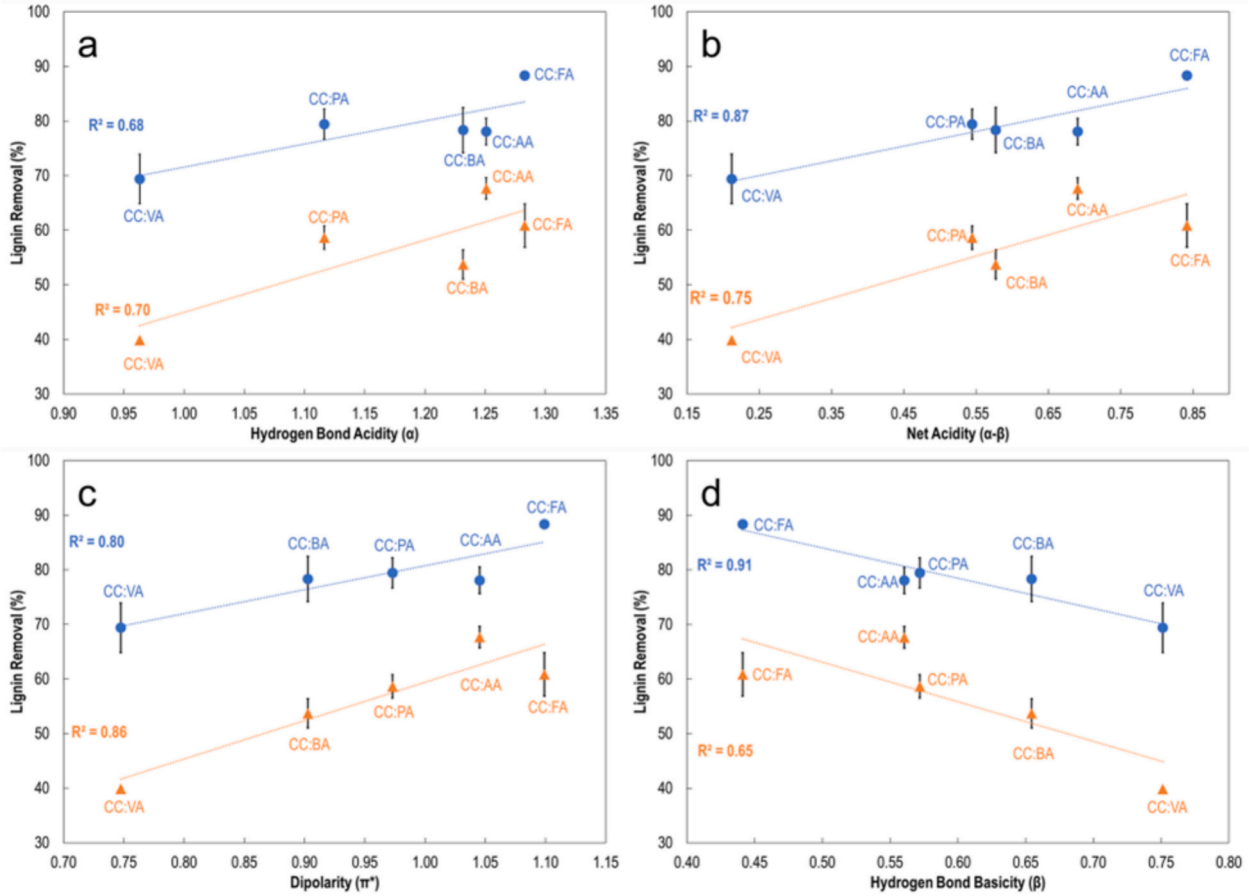


Fig. 3. Lignin removal from aspen (blue) and pine (orange) plotted as a function of the Kamlet-Taft α (a), α-β (b), π* (c), and β (d) parameters.

had a limited hydrogen bond accepting ability compared to the DESs more prevalent in other studies. As a result of the limited range of β values, no correlation between xylan removal and β was observed (Fig. 4a).

Similarly, no correlation between xylan removal and α was observed in the present study (Fig. 4b). Liang et al. [20] used a set of lactic acid-based DESs with α values ranging from 0.6 to 1.0 to treat corn stover and reported a positive linear correlation of xylan removal with α. The DESs with high α values had an enhanced ability to donate protons during DES-xylan interactions, allowing for cleavage of glycosidic linkages between xylobiose units. DESs with low α values have a relatively weaker capacity to donate protons when interacting with xylan, resulting in reduced xylan removal via glycosidic linkage cleavage. This is the reason behind the reported positive correlation between xylan removal and α when comparing DESs with high and low α values. Importantly, the α values of the DESs in the present work ranged from 0.9 to 1.3 which demonstrates that they were all highly acidic while occupying a relatively narrow range of α values. As a result, all five DESs have a strong ability to donate protons when interacting with xylan, hence the high xylan removal observed for each DES. Additionally, the fact that the α values of these DESs are similar to one another caused the xylan removal achieved by the DESs to be similar. Consequently, no correlation could be observed between xylan removal and α in this case, suggesting that there is a plateau in xylan removal efficiency for strongly acidic DESs. That xylan removal was high for all 5 DESs with little variation between them suggests that the acidity was high enough in each case such that differences in HBD alkyl chain length did not have a significant effect on the xylan removal efficiency within the scope of the selected DESs.

Arabinan and mannan, which also account for a significant portion of wood composition, may play a role in determining DES treatment performance. The presence of galactoglucomannans and arabinoxylans in both wood species is well known, though the extent to which DES pulping has an effect on their composition has not been adequately explored. Fractionation of arabinan and mannan was similar for both aspen and pine, ranging from 50 to 80 %. The strongest correlation was observed with arabinan and mannan fractionation from aspen and decreasing β values (Fig. 4c). Pine showed similar behavior though with weaker correlation strength ($R^2 = 0.59$). Additionally, a positive correlation was observed with the net acidity for both aspen and pine (Fig. 4d). As discussed previously, DESs with greater acidity are able to donate protons more effectively, allowing for hydrolysis of glycosidic linkages between carbohydrate units. It therefore follows that greater dissolution of arabinan and mannan was associated with greater values of net acidity for the selected DESs. Fractionation of arabinan and mannan due to acidic DES pulping has been reported previously, though to a lesser extent due to lower pulping temperatures [44,45]. It is possible that the severity of the pulping conditions in the present work facilitated the enhanced removal of these hemicelluloses.

3.5. Analysis of recovered lignins

After treating aspen and pine with CC:FA, the lignin extracted by the DES was recovered via precipitation and characterized by 2D-HSQC NMR and GPC. CC:FA was chosen for further analysis due to its strong performance with both aspen and pine. Compositional analysis of the resulting pulps revealed that performance metrics of the modified experiments mirrored those of initial experiments. CC:FA achieved 75.3 % lignin removal and 70.9 % hemicellulose separation from aspen and 54.3 % delignification and 46.2 % hemicellulose fractionation from

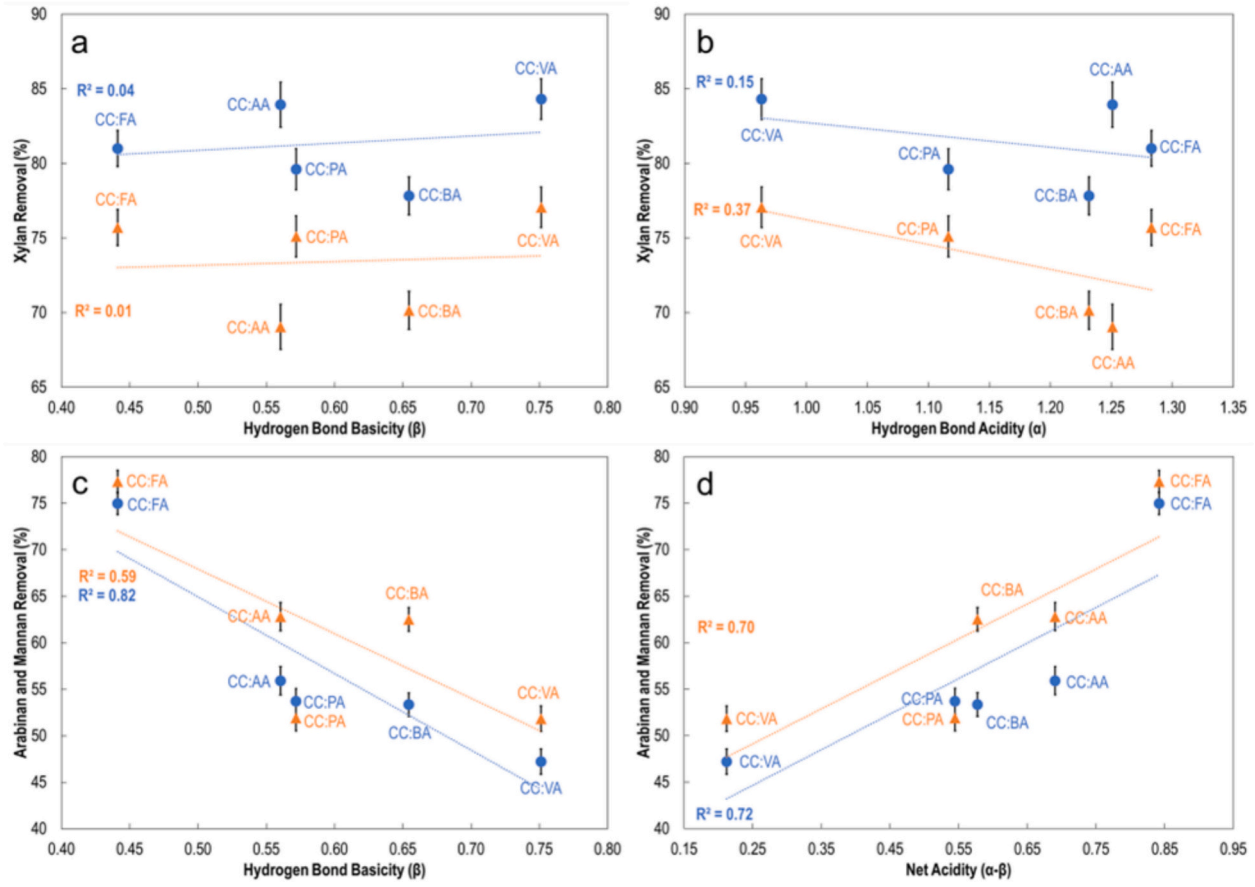


Fig. 4. The relationship between Kamlet-Taft α, β, and π* parameters with xylan removal and arabinan and mannan removal from aspen (blue) and pine (orange).

pine. Acetylation of the recovered lignins was confirmed by Fourier transform infrared spectroscopy (FTIR). Prior to acetylation, both samples exhibited broad peaks centered at 3300 cm^{-1} due to O—H stretching frequencies (**Fig. S4**). This peak was not observed for the acetylated lignins, which exhibited a new peak at 1765 cm^{-1} which was attributed to C═O stretching of acetyl groups (**Fig. S5**). The solubility of the acetylated lignins in DMSO-d_6 was significantly improved after acetylation, allowing for successful 2D-HSQC NMR analysis.

The yield of precipitated material on the basis of lignin in the untreated feedstock was 76.5 % and 84.6 % for aspen and pine, respectively. These yields are elevated compared to the lignin removal measured for each pulp, indicating that the lignin precipitation process was not 100 % selective towards lignin and may have resulted in co-precipitation of other materials solubilized during DES pulping. Cross signals typically associated with xylopyranosides and mannopyranosides were not observed in either the aspen or pine spectra [46], therefore precipitation of extracted hemicelluloses was not prevalent, though these species may have been separated during the acetylation process. Further analysis of the behavior of solubilized biomass components during lignin precipitation is required to better understand this phenomenon.

The aspen and pine lignin samples exhibited similar inter-unit linkages in the aliphatic side-chain region, with β-aryl ether (β-O-4), resinol (β-β), and phenylcoumaran (β-5) found primarily at δ_C/δ_H 60.6/3.19, 53.1/3.10, and 54.5/3.72 ppm. Signals corresponding to coniferyl alcohol were also identified at δ_C/δ_H 58.0/4.41 ppm. The full assignment of aliphatic NMR cross signals can be found in supplementary information (**Table S3**). Importantly, signals associated with the Hibbert ketone (HK) were identified in both samples at δ_C/δ_H 44.7/3.68 ppm (Fig. 5). The HK is a structure characterized by a ketone functional group on the β-carbon of the lignin sidechain and has been found in lignin recovered from organosolv and ionic liquid treatments [33,47]. The presence of the HK is cited in the literature as evidence of acid-catalyzed cleavage of β-O-4 ether linkages [33,47,48] which has been posited to be the predominant mechanism of delignification during DES pulping [3,40]. This mechanism has been well documented as one of the primary pathways of lignin acidolysis reactions [49,50]. Protonation of the hydroxyl group on the α-carbon of a β-aryl ether structure yields a benzyl carbocation intermediate. Once formed, this intermediate may undergo nucleophilic attack by other lignin fragments, yielding recalcitrant condensation structures. The intermediate also may experience enol-ether tautomerization which results in the release of guaiacol, and formation of HK as depicted in **Fig. S2** [48,51]. The presence of HK in both aspen and pine recovered lignin samples is consistent with these findings, indicating that β-O-4 ether cleavage played a significant role in lignin removal for both feedstocks.

The aromatic regions of the NMR spectra showed that the aspen lignin was rich in both guaiacyl (G) and syringyl (S) units, whereas the pine lignin was composed primarily of G units (Fig. 6). The full assignment of aromatic NMR signals is listed in supplementary information (**Table S4**). Signals corresponding with syringyl $C_{2/6}$-$H_{2/6}$ appeared at δ_C/δ_H 105/6.62 ppm while guaiacyl C_2-H_2, C_5-H_5, and C_6-H_6 correlations were found at δ_C/δ_H 112.7/6.98, 118.1/6.91, and 122.8/7.02 ppm, respectively. Additionally, $C_{2/6}$-$H_{2/6}$ correlations for syringyl and guaiacyl units bearing oxidized α-carbons were observed in the aspen spectrum at δ_C/δ_H 106.4/7.29 and 110.2/7.38 ppm, respectively. A cross signal attributed to p-coumarate was detected in the aspen lignin at δ_C/δ_H 131.3/7.76 ppm, indicating that a small portion of p-coumaryl (H) type lignin units were present as is typical for some hardwood species. Similarly, p-benzoate signals were detected in both samples at δ_C/δ_H 136.3/7.76 ppm which are commonly observed for both feedstocks [52].

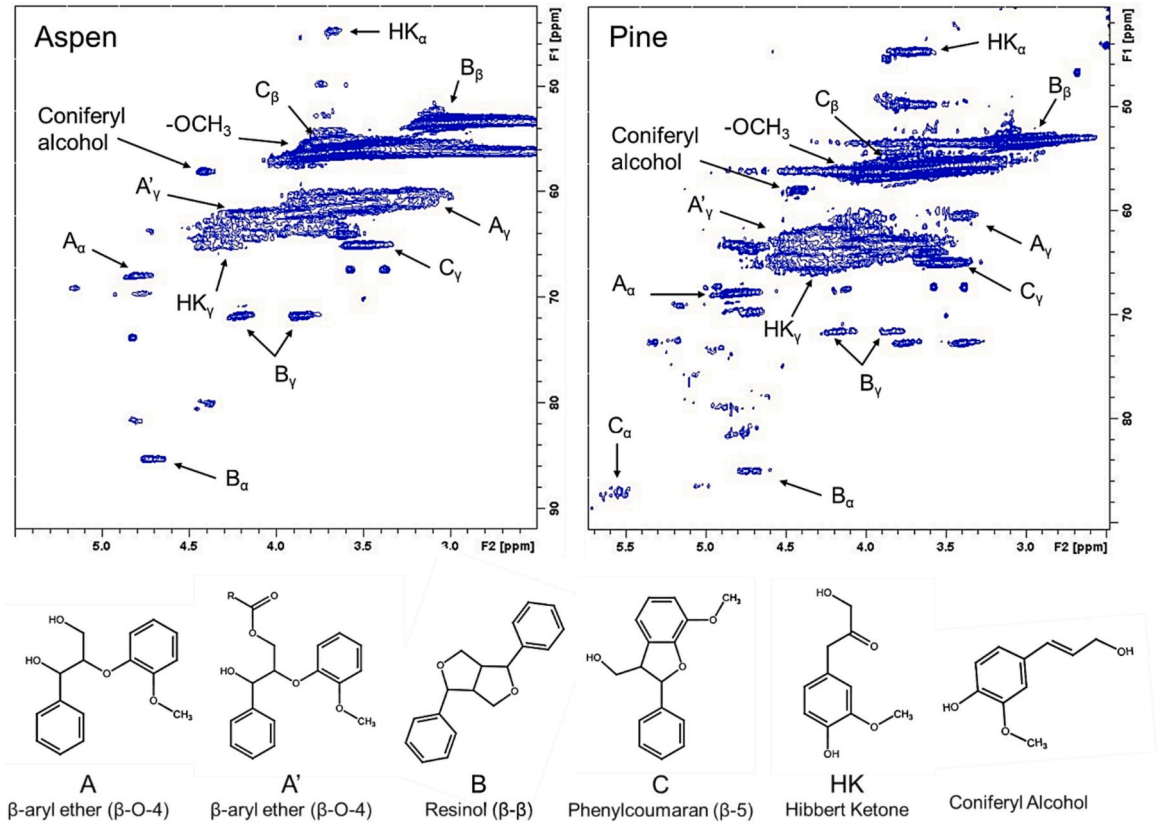


Fig. 5. Aliphatic sidechain region of aspen and pine recovered lignin analyzed by 2D-HSQC NMR.

As discussed, hardwood lignin containing syringyl and guaiacyl units typically has greater ether linkage content than softwood lignin containing primarily guaiacyl units [24]. It has been suggested that DES-mediated lignin extraction is more effective in treatment of hardwoods due to the susceptibility of ether linkages to DES attack [3]. This, in turn, suggests that syringyl units in hardwood lignin may be more easily removed during DES pulping of hardwoods. Interestingly, the S/G ratio of the aspen lignin was 1.20 based on integration of syringyl and guaiacyl signals. Native aspen wood lignin is known to have a higher S/G ratio between 1.7 and 1.8 based on analysis of milled wood lignin [31,53]. If syringyl units were much more receptive to DES attack than guaiacyl units, then the lignin recovered from the filtrate should contain a much greater fraction of syringyl units than guaiacyl units. However, the S/G ratio is closer to 1:1, meaning that the lignin extracted by CC:FA contained similar proportions of syringyl and guaiacyl units. This is corroborated by the presence of Cα-oxidized syringyl and guaiacyl units in the aspen NMR spectrum which indicates that DES attack occurred at both sites. Therefore, CC:FA did not display strong selectivity towards syringyl units during DES treatment of aspen.

Treatment of pine with CC:FA at 130 °C resulted in much lower lignin removal than aspen, mirroring the results of the pulping trials conducted at 150 °C. The reduced ether linkage content of softwood lignin may have contributed to this disparity, offering fewer reactive sites within the lignin matrix for DES attack. Additionally, the greater degree of condensation and cross-linking which is characteristic of native softwood lignin may present a physical barrier to the diffusion of the DES into the biomass substrate. This would prevent the DES from fully penetrating into the feedstock, denying access to certain fractions of lignin. A comprehensive analysis of DES diffusion rates through different biomass substrates may provide a more complete understanding of this effect.

Molecular weight averages of both lignins are listed in Table 3. Overall, the molecular weight distributions were quite similar, with pine having a somewhat broader distribution as evidenced by its polydispersity index of 2.0 compared to 1.7 for aspen. This striking similarity may have resulted from the lignin precipitation process which may only be capable of precipitating lignin of certain molecular weights. The supernatant obtained after lignin precipitation shows a brown discoloration (**Fig. S1**), suggesting that smaller molecular weight lignin fragments may not have been precipitated. The number-average molecular weights (M_n) are very similar, however the weight-average molecular weight (M_W) of the pine lignin is larger than that of aspen. This suggests that larger molecular weight fragments of the pine lignin were solubilized by CC:FA during pulping [21,33]. Additionally, when compared to milled wood lignin (MWL) and cellulolytic enzyme lignin (CEL) extracted from aspen and pine which represent the native lignin structures, the lignins recovered after DES pulping have much lower molecular weights. This indicates that significant lignin depolymerization occurred during pulping with both feedstocks.

4. Conclusions

The Kamlet-Taft solvatochromic parameters of deep eutectic solvents (DES) comprised of choline chloride and 5 saturated monocarboxylic acids were measured. Increased alkyl chain length resulted in decreased acidity (α) and net acidity (α-β), a corresponding increase in basicity (β), and decreased solvent polarity as shown by the decrease in π* and increased E_T. Biomass treatment experiments showed that lignin and hemicellulose separation was greater for aspen than pine, with lignin removal showing correlation with α, β, and π* for both feedstocks. Xylan fractionation was high for both aspen and pine with little variation between DESs. As a result, no correlation was observed with the α parameter, with similar performances here are attributed to the high acidity of all studied DESs. Removal of arabinan and mannan was found to correlate with decreased basicity and increased net acidity.

2D-HSQC Nuclear Magnetic Resonance (NMR) analysis of recovered

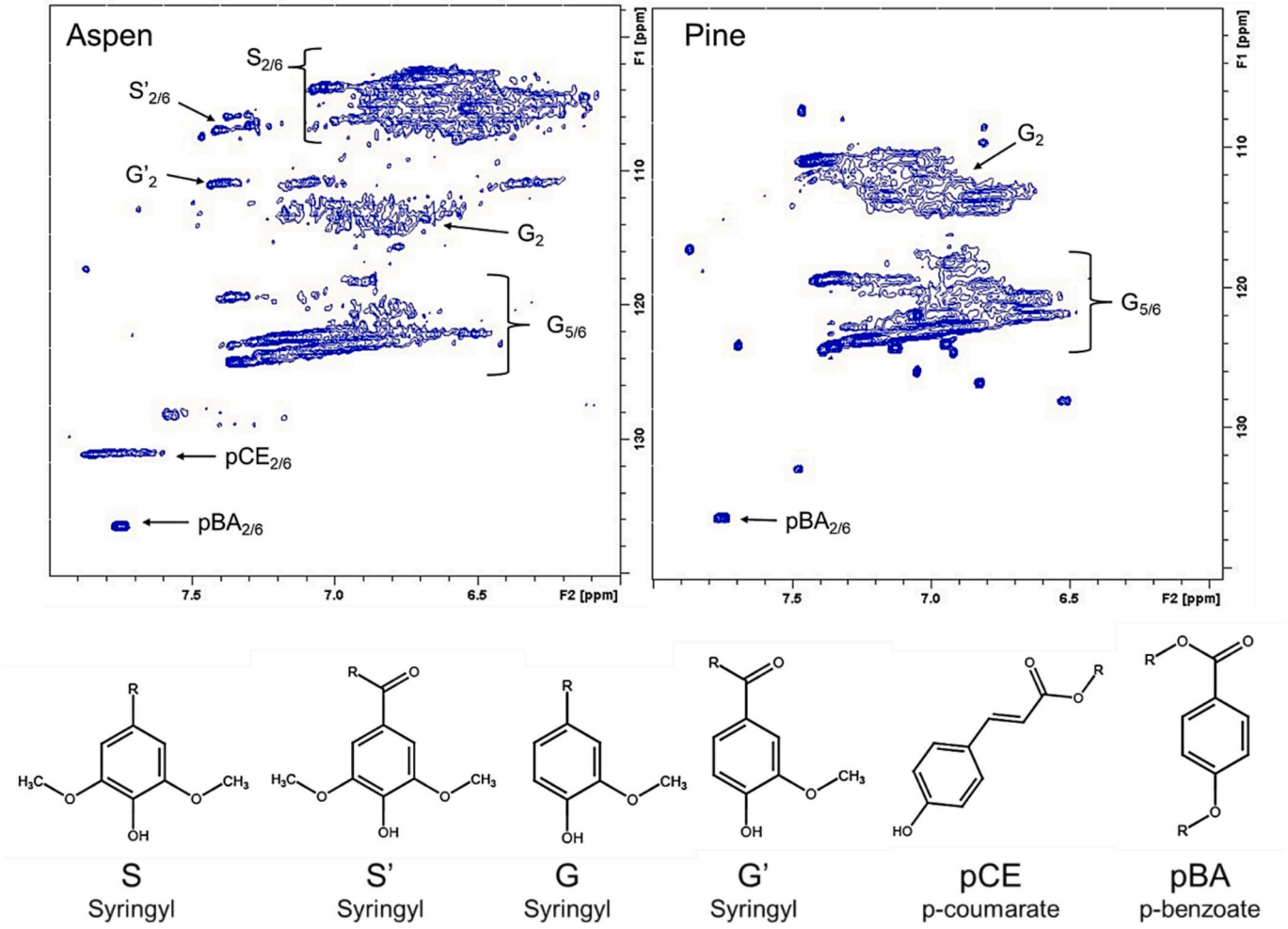


Fig. 6. Aromatic region of aspen and pine recovered lignin analyzed by 2D-HSQC NMR.

Table 3
Molecular weight averages of recovered aspen and pine lignin after DES pulping and representative native lignin samples.

	Mn (g/mol)	M_W (g/mol)	PDI	Reference
Aspen-DES	1500	2600	1.7	This work
Pine-DES	1500	3100	2.0	This work
Aspen-MWL	3800	6000	1.6	[33]
Pine-CEL	7200	12,000	1.8	[21]

lignin demonstrated that cleavage of β-O-4 ether linkages was the primary mechanism of delignification during DES treatment for both feedstocks, though selectivity towards syringyl units was not supported. Gel Permeation Chromatography (GPC) analysis confirmed that both aspen and pine lignin were significantly depolymerized by the DESs. The similarity between the molecular weight distributions of aspen and pine lignins may have been a consequence of the lignin precipitation process. The findings of the present work provide evidence that as favorable DES families are identified for applications, analysis of the DESs within the family must be conducted, since trends in performance may deviate from broader trends reported in the literature. The extent to which strong relationships can be established is also dependent on the type of biomass feedstock which adds further need for study in this area regarding the physical properties of the biomass feedstocks to elucidate their influence on DES delignification efficiency.

CRediT authorship contribution statement

John H. Sanders: Writing – original draft, Investigation, Conceptualization. **Phoenix Tiller:** Writing – review & editing, Investigation. **Seong-Min Cho:** Writing – review & editing, Investigation. **Richard Venditti:** Writing – review & editing, Supervision, Conceptualization. **Sunkyu Park:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijbiomac.2025.146086.

Data availability

Data will be made available on request.

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